Nauman Ahad

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https://nahad3.github.io

SUMMARY

Recent Ph.D. graduate specializing in Machine Learning and Signal Processing, eager to apply expertise to solve real-world machine learning challenges. Proven experience tackling open problems across diverse multi-modal time series data, alongside a strong foundation in Signal Processing tasks like distribution estimation, filtering, sensing, and change detection in signals.

EDUCATION

Georgia Institute of Technology, Atlanta, GA

Ph.D. in Electrical and Computer Engineering — August 2024

Concentration in Signal Processing and Machine Learning
Thesis: *Detecting and leveraging changes in temporal data*
Devised new methods for detecting distribution shifts, active learning, self-supervised deep learning

Masters in Electrical and Computer Engineering — May 2018

National University of Sciences and Technology, Islamabad, Pakistan

Bachelors in Electrical Engineering — May 2014

SKILLS

Programming/Scripting languages: Python, C/C++, MATLAB, Bash
Machine Learning/Data Science tools: Pytorch, TensorFlow, Keras, Sci-kit learn, SQL, Pandas, SciPY,OpenCV
DevOPS tools: Docker Containers, Ansible, Git, Jenkins, Linux, OpenStack, AzureVMStack
High Performance Computing: OpenMPI

WORK EXPERIENCE

The Coca-Cola Company, Atlanta, GA

Applied Machine Learning Researcher (Fixed term contractual project) — January 2024 - August 2025

- Collaborated with a diverse team of physiologists and biomedical scientists on https://sweatratecalculator.com
- Developed novel personalized hydration prediction algorithms via self-supervised learning on wearable multi-modal data.
- Assisting with data trial collection and developed pipelines to process, filter and extract features from raw signals.

Georgia Institute of Technology, Atlanta, GA

Graduate Research Assistant — Aug 2018 - January 2024

- Devised new improved robust change detection methods for real-world noisy settings
- Devised new machine learning methods for deep representation learning, active learning, and domain adaptation

NEC Laboratories America, Princeton, NJ

Research Intern in Machine Learning — May 2021 - Aug 2021

- Devised new self-supervised learning methods for data efficient ordinal classification (labels with inherent order)
- These methods reduce training time by 60 % for ordinal classification
- Developed Signal to Noise ratio classification methods that achieve an accuracy of 95 % for real world optical network data
- Explored how soft-labels are better suited for classifying noisy ordinal data

xFlow Research, Islamabad, Pakistan

Software Engineer — Sep 2014 − June 2016

- Developed a Python benchmarking suite for quantifying cloud computing infrastructure performance
- Tested newly developed Network functions virtualization installation packages on cloud infrastructure for deployment

Ph.D. PROJECTS

Semi-supervised sequence classification through change point detection

- Devised a method to use detected unsupervised change points for obtaining similar dissimilar pairs
- These pairs provide weak supervision that can be used to learn neural network representations in a label efficient manner

WiSAT: An activity recognition system for wheelchair users

- Devised an interpretable classifier for characterizing in-seat movement for paraplegic wheelchair using pressure sensor data
- Assisted in embedding this classifier in iOS/Android applications for end users
- Assisted in devising data collection protocols
- This tracking system is deployed in clinical trials

Sparse metric learning for change point detection

- Devised a new method that uses available change points to improve change point performance
- This method obtained triplet pairs from available change points for learning a metric for Wasserstein distances
- Demonstrated improved change detection performance on human activity, sleep and neuroscience datasets

Collaborative filtering for recommender systems using categorical matrix completion

- Devised a recommendation system that obtained user movie ratings by setting up an ordinal matrix completion problem
- Demonstrated that decomposing a caterogical matrix completition problem into 1 bit low-rank matrix completition problems can help improve movie recommendation performance by upto 5 %

Domain adaptation for time series classification via selective channel screening

- Identified scenarios where it is useful to learning invariant feature representations between source and target domains
- Explored appropriate distance metrics in representation space to mine similar class samples across domains
- Designed new method that can ignore spurious and noisy channels that hamper classification performance in new, unlabelled domains. Improves performance over State of the art methods by up to 14 %.

Self-supervision and active learning for learning temporal representations.

- Explored methods for learning self-supervised representations for time series datasets that do not require augmentations
- Devised active label sampling schemes suited for temporal neural network representations for instance classification tasks

TEACHING AND LEADERSHIP EXPERIENCE

Georgia Institute of Technology

Graduate Teaching Assistant Aug 2020 - May 2021

Conducted office hours. Assisted designing and grading homework problems for the following graduate level courses

- ECE 6270: Convex Optimization
- ECE 6258: Digital Image Processing

SELECTED PUBLICATION AND PREPRINTS

1. **N. Ahad**, M. Davenport, " Semi-supervised Sequence Classification through Change Point Detection ", *AAAI*, 2021
2. **N. Ahad**, E. Dyer, K. Hengen, Y. Xie, M. Davenport, "Learning Sinkhorn divergences for Supervised Change Point Detection", *In revision, IEEE Transactions on Signal Processing. Preprint* arXiv:2202.04000
3. **N.Ahad**, S. Sonenbum, M. Davenport, S. Sprigle, " Validating a Wheelchair In-Seat Activity Tracker", *Assistive Technology*, 2021.
4. **N.Ahad**, M. Davenport, Y. Xie, " Data Adaptive Symmetrical CUSUM ", *Sequential Analysis*
5. **N. Ahad***, N. Nadagouda*, E. Dyer, Mark Davenport, "Active Learning for Time Instant Classification", *Workshop on Data Centeric Machine Learning Research, Int. Conf. on Machine Learning (ICML)*, 2023
6. **N. Ahad**, Mark Davenport, E .Dyer, "Time Series Domain Adaptation via Channel-Selective Representation Alignment", *to apper in Trans. on Machine Learning Research (TMLR)*, 2025
7. F. Zhu, A. Sedler, H. Grier, **N. Ahad**, M. Davenport, M. Kaufman,A. Giovannucci, C.Pandarinath " Deep inference of latent dynamics with spatio-temporal super-resolution using selective backpropagation through time ", *NeurIPS*, 2021
8. J. Quesada, L. Sathidevi, R. Liu, **N. Ahad**, J. M. Jackson, M. Azabou, J. Xiao, C. Liding, M. J., C. Urzay, W. Gray-Roncal, E. C. Johnson, E. L. Dyer " MTNeuro: A Benchmark for Evaluating Representations of Brain Structure Across Multiple Levels of Abstraction ", *NeurIPS Datasets and Benchmarks Track*, 2022

PATENT APPLICATIONS

- Mizoguchi, L. Tong, Z. Chen, W. Cheng, H. Chen, **N. Ahad**, "Ordinal Classification through Network Decomposition", US Patent App. 17/896,747 , 2023
- L. Tong, T. Mizoguchi, Z. Chen, W. Cheng, H. Chen, and **N. Ahad**, "Semi-supervised framework for efficient time-series ordinal classification", US Patent App. 18/152,238, 2023